

Moving Towards Mobility Justice: Challenges and Considerations for Supporting Advocacy

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In response to climate change and continued urbanization, urban transportation systems around the world are undergoing transitions to promote lower-emission vehicles, public transit, biking and walking. However, mobility is a complex issue that raises important questions of social justice as a result of its connections to numerous aspects of everyday life and the broader social and political contexts. Drawing on interviews with transportation advocates across Canada, we identify four ways in which the design and use of existing mobility tools and technologies perpetuate mobility injustices, and deepen the divide between urban planners and the public. Looking across these arguments, we note path-dependence in transportation knowledge infrastructures as a common barrier to mobility justice advocacy that can be difficult to recognize or overcome. Finally, we consider tactics that research in HCI and CSCW might pursue as part of efforts to unsettle path-dependence and reorient transportation planning towards mobility justice.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**.

Additional Key Words and Phrases: digital advocacy, transportation, mobility, data practices, digital civics, civic engagement, lockin, path dependence

ACM Reference Format:

Tanee S Agrawaal, Samar Sabie, and Robert Soden. 2024. Moving Towards Mobility Justice: Challenges and Considerations for Supporting Advocacy. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1, Article 96 (April 2024), 22 pages. <https://doi.org/10.1145/3637373>

1 INTRODUCTION

The ongoing climate change crisis is having a significant impact on transportation planning and policies for urban metropolises. Transportation systems contribute significantly to greenhouse gas emissions, a major contributor to climate change. As a result, there is a growing recognition of the need to reduce these emissions in order to mitigate the negative impacts of climate change [64]. In North America, for example, governments, urban policy planners and municipalities have started to develop emissions reductions plans that electric vehicles and alternative modes of transport such as public transit, biking and walking [2, 3, 40]. They are also evaluating the resilience of transportation systems in the face of extreme weather events, such as floods, which are expected to become more frequent and severe as the planet warms [82]. In addition to the environmental impacts, the dominance of automobility in transportation planning over the 20th century has also led to consequences which include detrimental health impacts, congestion, long commutes and social segregation [29, 30, 83]. This has prompted many metropolises towards NetZero climate

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2573-0142/2024/4-ART96 \$15.00

<https://doi.org/10.1145/3637373>

strategies [71], VisionZero safety policies [11], transport demand management and an overall transition in urban mobility systems and technologies.

HCI and CSCW have pursued a number of areas of study relevant to such a transition. Researchers have studied the impact of digital tools on the organizational practices of civic groups [6, 72], the design and use of computing technologies in participatory processes between urban planners and the public [8, 9, 36, 45], the role of information and communication technologies (ICTs) in community engagement and accessibility challenges of the current transportation infrastructure [48, 80]. Other researchers have studied the advantages of supporting data infrastructure literacy – combining sensibilities for the socio-technical infrastructures that datasets interact with in addition to just technical capacities – in empowering advocacy work [39]. What this body of work shares is a recognition that, depending on its constitution and use, knowledge infrastructures can play an important role in either aiding or hindering advocacy efforts. While this research offers important insights on how to ensure the former, urban mobility remains an exceedingly complex issue and connects to a myriad of social and political factors that influence freedom of movement.

Research has the numerous connections between mobility and social justice [85]. This includes notable insights into how transportation systems and infrastructures significantly impact people's ability to access education, employment and other essential resources, therefore, playing a role in shaping social and economic opportunities [29, 84]. Over the long history of cities, movement has been politically charged – it has been free for some and heavily controlled for others, dividing people into the “kinetic elite” and “mobility poor” [53, 58]. This division has resulted in increased risk of various harms of mobility like health impacts due to transport-related pollution, and risks to bodily safety on the streets, to mobility-vulnerable groups like women, children, and disabled people [77, 94]. Mimi Sheller argues for recognizing and addressing such socio-political divides through mobility justice, which is “a systematic framework that analyses multiple interacting relations of movement – for example, people, goods, information, resources and capital” [85]. This paper aims to study mobility justice activities in Canada and highlight the challenges that advocates face with using current movement-related information, data and tooling in their day-to-day advocacy work.

We identify four main challenges that mobility justice advocates face when interacting with transportation technologies and in advocating for mobility justice. First, we argue that the design of existing mobility tooling and information sources deepen the divide between urban planners and the public. Specifically, these divides are created by the high technical expertise required for transport-related proposals and information that is detached from the local community's language and needs. Secondly, we find that advocates often have to build additional data and tooling for funders and financial grant reports. This creates immense amounts of additional labor for resource-constrained advocates in building new skillsets. Thirdly, our interviews suggest that the planners' focus on innovation in planning urban mobility technologies and infrastructures limits their investment in well-known solutions. Finally, our fourth finding highlights the overall resistance of the planners, and public to change the current mobility landscape. As we will show, these findings shed light on how planning, information and decision making infrastructures exert significant path dependence that can perpetuate mobility injustices and are relevant to several ongoing areas of research within HCI and CSCW.

In the next section, we outline data practices and data infrastructuring for empowering advocacy work. We then proceed to define and include relevant literature on mobility justice and the various tools and technology systems within HCI that have been built for supporting mobility justice advocacy. We then provide a short overview of our participants – the Canada-based mobility justice advocates we interviewed – and our research methodology for this paper. In the Findings section, we draw on data from our interviews to outline the four ongoing challenges and questions that these advocates are grappling with, and the strategies they employ to overcome those challenges. Finally,

in the Discussion, we raise the significant influence of “path-dependence” [60] in transportation politics and the knowledge infrastructures that inform them. We then offer recommendations for HCI and CSCW researchers and designers to engage with issues of path dependence and contribute to mobility justice. This paper thus contributes insights into current technological practices in transportation and its challenges. Furthermore, we highlight tactics employed by mobility justice advocates to overcome those challenges, and offer recommendations for HCI and CSCW research to contribute towards mobility justice, related data practices and policies.

2 LITERATURE REVIEW

2.1 Data, Data Practices and Data Infrastructuring for Advocacy work

As with other areas of social and political life, the burgeoning data-driven decision-making movement is also gaining traction within non-profit and other advocacy organizations. For mission-driven organizations, this adoption is often a result of heightened pressure to provide performance data to funders, clearly demonstrated in their adoption of information management systems, business intelligence tools and various other data analysis activities. Recently, scholars [23] have examined the challenges that exist within non-profit advocacy work through data feminism [27], a concept and framework that explores the intersection of data science and feminism. While increased data-driven practices tend to improve efficiency and productivity for non-profits, they contribute significantly to complex digital divides [56]. Implementing tools and services that support these practices requires that organizations invest scarce resources like time and money into building computational literacy [16, 49, 63]. Given that a one-size-fits-all approach doesn’t work in this context, individualized tooling requirements for different projects and contexts can exacerbate complexity when compared to more standard use cases. Verma and Volda find that even leading organizations are unable to sustain such practices in the long-term due to significant data warehousing challenges, a high amount of manual data entry tasks, and siloed data [93]. Bopp et al. find that if not implemented carefully, information systems can propagate a cycle of increasing disempowerment for mission-driven organizations that ultimately impedes rather than improves nonprofit work [15].

Prior HCI literature has used the framing of “publics”, formed through a shared set of social conditions [25], to understand advocacy and the link between multiple stakeholders, issues and attachments across boundaries [8]. In designing for data infrastructures that can support advocacy across multiple publics, researchers have argued for the importance of the relation between data infrastructures and their publics which is visible in their continuous interaction, influence and reshaping of each other [39]. Recognizing this relationship in digital advocacy work requires that advocacy work and its workers be able to not only understand data but also data sensibilities. Gray et al. have termed this as “data infrastructure literacy” that encompasses the human and non-human actors in data-related activities including the people, groups, software, hardware, social, cultural and political contexts in which data are created, analyzed and utilized [39]. As we will discuss, data infrastructure literacy, in its sensitivity towards the alignment of data infrastructures towards varied interests, provides a valuable framework to support the multiple modalities of urban mobility justice work.

Researchers have studied the shaping forces of socio-spatial infrastructures on human and collective experiences in light of increased “datafication” in public discourses and policymaking about community engagement [28]. With increased sensing capacities, data has found to be enmeshed with communities and their social geographies through a myriad of data infrastructures such as public urban displays and energy monitoring systems [13, 51]. This link of data and communal life has led researchers to examine the reconceptualization of data and its continual reenactment across

networks of people, place and technologies, being termed as “data-in-place” [90]. Through their engagements with these issues, Taylor et al. examine the different functions of data in during its development and use, identifying how it becomes bound up with, and contributes to the character, of place. They argue for design systems to preserve the situated qualities of data to capture and enable the relational configurations of data, community and place.

2.2 Mobility Justice

Mimi Sheller defines mobility justice as a relational bridge between the triple crisis of urbanization, migration and climate as they all can be linked to crises of mobility [85]. For example, fossil-fuel dependent transport is a major contributor to climate change, and it is driving a series of climate mitigation actions and policies to reduce greenhouse gas emissions, reform land-use patterns and transform transport infrastructures [10]. Secondly, the increasing urbanization in cities across the globe relate to a spread of “automobility” which deepens issues such as hazardous pollution and social isolation, and also results in poor road safety and disparate movement accessibility [30]. While research suggests that these urbanization crises are being approached through innovative technologies such as smart mobility [31], vehicle sharing [68], and Vision Zero policies [11] among others [24, 66, 69, 95], scholars argue for more transformative mobility systems that address underlying causes of a “splintered urbanism” [38]. Lastly, the global migration crisis over closed borders and restricted cross-country movements have resulted in a strong divide between movement capacities. In relating the refugee crisis to climate change and urbanizations, crisis and disaster researchers have studied uneven mobilities as a consequence of post-disaster activities such as land grabs, terming the whole phenomenon as “disaster capitalism” [50].

In understanding how mobility injustices traverse issues, mobilities research seeks to define mobility justice at macro and micro-levels of scales of movement. At the macro-level, beyond transportation in cities, researchers have closely examined social parameters such as race, class, gender, disability and sexuality that affect micro-mobility processes [17, 41, 82, 83, 86]. Studies have also shown how these effects have manifested in cross-border issues like the refugee and migrant crises, and in matters relating to global tourism and immigration [84]. At the micro-scale, studies have investigated common narratives around “rights to the city” and the relation between transit justice and spatial justice [14, 29, 91]. For example, skewed urban transit planning, which often leaves out the safety needs of women, can result in the perpetuation of gender inequalities, and general inaccessibility [74]. Theresa Enright highlights similar goals between movements of mass transit and social justice and equity, calling attention to the role of transit infrastructure in shaping contemporary urbanization including affordable housing, and urban poverty reduction [29]. The evident complexity of urban mobility regimes makes linear technological solutions and policies unable to sufficiently address issues of mobility justice. Instead, studies call for mobility justice as a relation between human and non-human entities, a local and global concern that views mobilities as continuously contested, contingent and performative [29, 85].

Another theme that we engage with through this consideration of path dependence is the historical politics of mobility and its related technologies, which has defined the circulation and placement of people, information and resources. Liberal governances of mobility bring the concept of “othering” insofar as free movement is concerned. Hagar Kotef describes “free movement of some [people]” as one that “limits, hides, even denies the existence of others” [53]. Hobbes and Locke describe a split between the movements of the citizen groups (whose movement should be maximised) and other(ed) groups (whose movements are considered as disruptions and dangerous) [7, 58]. An analysis of borders and security policies in Europe, the United States and other white countries reveal the tight regulations over movements of people of color quite clearly. The rise of recent social movements that call attention to these interconnected problems of mobility justice,

including the Black Lives Matter ¹ and the Free Transit Toronto ² initiatives, has been encouraging [29]. However, scholars advocate for mobility justice systems that share common goals with social justice movements but “genuinely redistribute life chances and well-being across the full range of urban populations” [85].

2.3 Digital Advocacy and Activism for Mobility Justice

Mobility justice advocates undertake a myriad of activities to enable and support policy changes around issues of mobility like campaigning, supporting community-led initiatives or collaborating with other mobility justice organizations, advocating around issues like road infrastructure inaccessibility, public transit affordability or transport sustainability, and working with stakeholders like non-profits, individual civic groups, and urban planners or municipal departments. Recently, there has been an increased focus within the HCI community towards building novel and creative digital tools or platforms to support mobility justice advocacy efforts. For example, Kirkham et al. used video footage and GPS-based evidence for reporting road inaccessibility, specifically for wheelchair users [48]. Other studies focused on assisting local government officials with infrastructural inaccessibility include the IBM Sidewalks, and ‘chairable’ technology designs that use passive sensing for wheelchair users. Mobile technology for mobility justice activism has also explored user-generated citizen journalism technologies for recording dangerous road behaviour [1, 4], and peer-surveillance for civic-transport activism. These peer-surveillance or ‘sousveillance’ approaches utilize user-generated mobile content [62], like live-streamed video content for “bottom-up” activism [19, 54], and question the information asymmetry of “top-down” surveillance approaches [32]. However, concerns about the motivations and trust about contributors towards citizen journalism remain unresolved [34].

While these tools offer remarkable design insights into supporting digital advocacy for mobility, they tend to rely heavily on user-reported data and evidencing. These approaches tend to overlook the challenges surrounding the curation of evidence itself, leaving the issue largely unresolved [8]. In recent years, digital advocacy and HCI scholarship have shifted focus towards building technologies that situate mobility justice advocacy as important sites of socio-political change [8, 45, 77, 80]. In studying cycling advocacy efforts in two US metropolises, Asad et al recommend that designers focus on building tools or platforms that support citizen dynamism in their attachment to issues, and that support advocacy across multiple public identities [8]. Similarly, researchers have found that encouraging new data formats and shifting communication of issues away from traditional formats in advocacy [55, 76], designing for adaptability and being mindful of the diverse perspectives that exist in mobility will enable better addressal of issues within public transport [77, 80]. More generally, researchers have also proposed a playbook for designing city-scale systems that support community engagement, which is at the core of digital civics research, through interactions between city officials and the general public that are relational instead of transactional ones [9].

HCI and CSCW researchers have also investigated approaches to support more cooperative and effective interactions between the wide variety of stakeholders involved in urban planning [26, 35, 72, 94]. Recognizing the importance of disparate motivations, desires and perceptions of these stakeholders, digital civics research that is focused on advocacy has sought to conduct multi-stakeholder analyses in its approach. One such study has found that urban issues are often reflective of broader patterns of social and political inequity [80]. Their study demonstrates that obscure accountability structures within urban planning may often cause conflicts between the urban citizens and governmental organizations and there is a requirement to render these accountabilities

¹<https://blacklivesmatter.com/>

²<https://www.freetransittoronto.org/>

Table 1. Interview participants' background and experience

#	Role and Background	Experience (in yrs)
1	Senior Director of a transit justice organization	12 years
2	Executive Director of an active transportation organization	10 years
3	Campaign Manager for transportation advocacy and Host of transit podcast	10 years
4	Senior Analyst for transportation in an environmental justice organization	5 years
5	Project Manager for mobility justice organization	8 years
6	Co-founder and Strategy and Design Lead of an urban non-profit	10 years
7	Independent Transportation and Housing Justice Advocate	5 years
8	Advisor and member of five transit justice organizations	20 years
9	Research Manager at environmental justice organization	9 years
10	Research Associate at environmental justice organization	5 years
11	Community Director for environmental justice organization	11 years
12	Lead for Toronto Chapter at environmental justice organization	6 years
13	Program Manager for national environmental organization	7 years
14	Lead Coordinator and Program Designer for sustainability organization	4 years
15	Program Manager for environmental justice organization	5 years
16	Program Manager for environmental justice organization	2 years

and responsibilities visible. Furthermore, they stress on the key need to improve government feedback and include the local community's input in planning decisions that affect them the most. In designing for better means of gathering the community's input, Le Dantec et al. find crowdsourced data particularly useful, and demonstrate the three roles these data can undertake – as authority, as evidence and as ambivalent [55]. Returning to the playbook, Asad et al. recommend creating technologies that accommodate mutual learning opportunities for all stakeholders in order to encourage more participatory forms of urban planning [9].

3 RESEARCH METHODS

This research is based on a qualitative examination of mobility justice activities, discussions, challenges and solutions associated with in urban cities of Canada. The data presented in this paper is drawn from 16 semi-structured interviews conducted by the first author between June and September of 2022 with mobility justice advocates who are associated with a diverse range of organizations across the country (See: Table 1). Participants were recruited through civic technology interest groups, snowball sampling, google searches and some personal connections. Of the 16, eight interviewees identified as being part of explicitly mobility justice organizations, and the remaining eight worked on topics related to mobility justice but had focuses on other aspects like sustainability. Nine interviewees were based in the Greater Toronto Area (GTA) with operations in neighboring cities like Mississauga and Markham, six interviewees came from national organizations that worked in multiple cities like Montreal, GTA and Vancouver, and one was from London, Ontario. In general, our interviewees were senior figures in the organizations they were part of, most of them were program managers leading projects, principal coordinators for the organizations' work in a city, or even chief executive officers for their organization.

Interviews lasted between 45 and 60 minutes, and explored the interviewees' experiences with active campaigning, and championing mobility justice in some of the major cities of Canada. The interviews were broad but designed to record day-to-day activities and primary projects undertaken by mobility justice organizations, technical tools and resources used and produced by them to aid

with their advocacy efforts, and challenges faced by the urban community. Other topics included how these organizations conduct planning sessions and meetings, the various stakeholders they interact with, and their outreach and recruitment strategies. At the time of writing, all of these groups remain active in mobility justice work. All our interviews took place remotely over Zoom and were video recorded. Recordings were transcribed using Otter, and the first and second authors read and revised the transcripts to verify the transcription prior to analysis. The data collection process for our study was conducted in line with institutional research ethical guidelines including a research ethics approval prior to the interview study. We obtained informed written consent from all participants, ensuring they were fully informed about the purpose, and procedures involved in the study. The collected data were securely managed on password-protected and encrypted platforms to safeguard the privacy and confidentiality of participants. Following relevant ethical guidelines and regulations, we treated participants with respect, dignity, and fairness and present anonymized and non-identifiable data in this paper.

After reviewing all the interview transcripts for errors, the first author and second author conducted a reflective thematic analysis on the interview dataset as described by Braun and Clarke [20], similar to prior CSCW work [46, 88]. Author A and author S took on the role of “coders” and assigned sections of the data to distinct codes. This light coding was conducted on Otter transcripts from which preliminary themes were developed by each author independently. These preliminary themes and their descriptions are provided in the supplementary materials. Since the interviews were broad in their design, the codes were iteratively refined into hierarchies via discussion sessions that spanned several meetings. During these meetings, codes from both coders were updated, refined and merged till distinct relevant codes and themes emerged. The codebook included eight codes: Information accessibility, Lack of holistic transportation planning, Challenges with data and digitization, Need for tracking impact, Increased focus on innovation, Changes required in people’s behaviour, Increased focus on business impact, and Fear of backlash. We provide the semi-structured interview questionnaire, and the final codebook used by the research team in the supplementary materials. The authoring team applied the updated codebook to the entire dataset to reveal different categories of themes and sub-themes. We then described each category with short narratives that captured each aspect of the mobility justice work undertaken by our participants, noted possible challenges that these advocates face in their work, as well as the strategies that they employ to overcome these challenges. These categories and narratives were iterated over in several successive rounds of discussions with all the authors. We eventually converged on the final set of patterns or themes in the data, that are presented as Findings in the following section.

4 FINDINGS

4.1 The design of urban mobility planning tools and information systems deepens the disconnect between urban citizens and urban mobility planners.

Organizations who we spoke with for this research frequently discussed the deep disconnect between mobility planners and stakeholders of public mobility infrastructure including the public and government organizations. Although most cities in Canada have public participation procedures that include city planning meetings and town halls, our interviewees attribute this disconnect to the format of the public meetings, inaccessibility of information related to mobility planning, and the challenges in the design of frequently used tools including maps or other models of transportation systems. This includes highly technical proposals that are provided by the urban planners facilitate the public participation process but also limit who can participate in an informed manner. While talking about the inaccessibility of the proposals, one of our organizers expressed this sentiment,

"Any technical area by definition is hard to get into if you don't have a certain level of knowledge or training or experience already. So that barrier to entry, turns it into, you know, a conversation that only some people are allowed to be in, and you know that you end up with a lot of people very frustrated by decisions [transit-related], because they weren't in that conversation and feeling like things [mobility infrastructure-related policies] are being imposed." - P7

In addition, our participants highlight that the language of everyday transit posters, flyers and resources are generally not representative of the community they intend to serve. Quoting one of the transit justice advocates we interviewed, "another problem with city information is that it's just in English when there's like 50 languages spoken in Toronto." Speaking of the challenge foreign languages present to a regular transit-rider, one of our participants said:

"I've seen people get very confused when the bus makes a turn they're not expecting. And I speak English, we could probably figure it out given enough time. What if you don't speak English? It's very, very confusing to you. I had been in other countries that don't speak English and tried to ride a transit system. And sometimes it's very easy because they understand this and use shapes and pictures to show people what to do." - P3

Similarly, the design of transit maps inhibits access to the transit systems as a whole and the information about public transit. For example, as one of our transit justice activists said:

"Folks will travel to Brampton to go work in a warehouse, that sort of thing to know how the different systems connect together. And it seems to me that different transit systems, they'll publish a great map of their own system. But don't tell you how to go anywhere else. You can say maybe that's not their problem. But it kind of is, because it's how people actually access your system. Right?" - P9

Every organization we spoke with discussed this challenge, which suggests the extent of these problems and the inequalities they create in urban settings. Some of them focus solely on simplifying the content of these proposals by reducing onboarding effort for people. They bridge this gap by providing the people with the knowledge to evaluate different claims from different providers (politicians). They also identified the need to be non-partisan in their work so that their focus is on "clear and accessible information that breaks through the knowledge gap barrier and helps to educate or inform." To address these access barriers, they elaborate on the key role of social media (Twitter) in increasing their own access to transit-related topics that need to be simplified:

"Twitter creates such an opportunity for different individuals with lots of different backgrounds to come together to talk about topics that are of interest. And so that's part of where we actually get research topics is what's everybody talking about? What are the questions that people have, when we see a politician make a claim about the TTC?" - P1

Further elaborating on how they gather resources for their own research to increase accessibility:

"Because when they're having arguments, they're pulling out facts, and they're pulling out links and resources, and all of that. We're not often as worried about the argument. But we love all the information, right? All the sources, because that could be useful for later for us to talk about a certain thing. And it's interesting, because you could have a big, big argument about a certain transit discussion. And everybody involved already knows lots of stuff. And they're just they're arguing about how to solve this problem. And that might not be something we need to be involved in at all, because a whole bunch of people are already engaged, and they already have information. It might be supremely technical. Right?" - P1

In addition to limited information access, the planners' hyper-focus on small sections of the street or only on specific sites was another critical element in this disconnect. Most municipalities count car traffic counts, seriously injured incidences, emergency services and nearby major institutions in mobility infrastructure projects. However, the map data and tooling that inform these considerations do not capture the situated nature of an urban space – its social and experiential aspects – which leads to a lack of holistic planning. An active travel advocate captures this perspective:

“Ideally, of course, they would do a lot more than that [just numbers put together], ideally, they would look at the importance of measuring like perceptions of safety, and comfort and belonging on streets. And again, that very much ties into like the social conditions. You know, it’s nice if from an engineering best practice perspective, a wide sidewalk and bike lanes are safe. But if someone doesn’t feel safe, either because, you know, they don’t feel safe riding with the child there, or they don’t feel safe because of street harassment or they don’t feel safe because the name of the street is named after some colonizer who committed genocide.” - P2

The incorporation of similar metrics for every street is problematic because put quite simply, every street is not the same in its location, build or in the people using the street. In our interviews, advocates also brought up strategies they regularly employ to overcome this challenge. An advocate who works primarily with youth discussed one such strategy in the form of “empathy exercises” that help with holistic planning:

“I think it’s important to help them realize that there are obviously people who are younger and older than them with different kinds of needs and in different kinds of neighborhoods. So, one of the tools that we use is using personas. So based on based on the population of the city, or of the neighborhood that we’re looking at, we develop these sorts of personas or avatars. And we develop several of these personas to help youth relate better relates to the different kinds of demographics in the area? And what kinds of needs and what kinds of considerations that they need to take into account.” - P6

That organizer further illustrates how some tools like Google Mind Maps facilitate these exercises:

“So we use the mind maps to quickly show them how they could put labels on and color code them use different icons, etc. And put together these assets maps, that they can then go back to later on to say, Oh, like this neighborhood. They have a lot of X and not a lot of Y. And how what can we do to address that issue?” - P6

4.2 Mobility justice organizations often have to undertake the additional labor of putting data and tooling in place to manage financial and volunteer resources.

With limited financial resources and volunteers, mobility justice organizations frequently struggle with finding grants and reporting on their activities to their donors. These reports, which require organizations' data and activities in detail, are often essential for the longevity of these organizations. This detailed reporting is especially taxing for smaller organizations because they do not have the technological tooling in place for such assessment. Even for bigger organizations that have some tooling in place, it is challenging to maintain the tracking system. In addition to tracking impact, mobility justice advocates are compelled to formulate and decide on data management practices, and data privacy policies for these reports, which lies outside their area of expertise. While most ultimately did come up with some degree of impact tracking, many of our interviewees described this as one of the biggest challenges in their work. This sentiment is captured by one of them:

“The biggest challenge they [organizations] have is to consistently maintain and manage any tracking system whatsoever. And so, they [organizations] often find themselves sort

of coming back six months later be like, “Oh, what did we do because we have to now talk about this”. And so just do maintenance and keeping the data up to date and keeping that thing is just it’s such a mental work.” - P13

Another interviewee, who is part of an organization that supports mobility justice projects, talked about an organization’s data policies being a concern for even the public:

“Organizations we speak to often tell us, “We have people [who the organizations engage with] asking how does it work, you know, like the data policy, when they open for example, our application, there’s always like, here’s how we’re going to use your data, blah, blah.” This is a question like, data governance and data policy that every partner [mobility justice organizations] have right now.” - P10

Donor requirements for impact tracking has led mobility justice organizations to make many technological choices, including the use of social media, rolling out digital surveys and campaigning, putting in place mechanisms to track their own activities, and creating data management protocols. The adoption of these strategies is geared towards sufficiently applying for grants, but this comes at a cost – advocates often have to engage in additional labor to invest time and resources in learning and implementing these technical data procedures. For example, one of our interviewees talks about how they have to look for external expertise for this task:

“We are engaged with a planning graduate student from X University, who this year is sort of looking at all the data we’ve gathered, to try and identify like, what, what trends we’ve seen, since we started. And also help us identify how we can better track or understand some of our longer-term goals because we want to through this program, is to encourage people to feel more confident and participated more regularly. And we haven’t really figured out how to track that well yet.” - P2

Organizations or social enterprises that support mobility justice advocacy are now also building guidelines and tools to support this requirement for data policies. We interviewed one such organization who highlighted a “learning management system” that offers such support:

“Part of the element inside this whole big project is the learning management system. And it’s really like a platform where people, they can click on it. And like, there’s different modules to learn more about, for example, smart cities, how to implement, for example, an open data policy, how to implement a privacy policy. So, it’s online. And that will be another great tool. It’s pretty, it’s pretty extensive. It covers many different topics.” - P4

The same organization is also working on advising advocates on how a “Data Trust” could help manage their data:

“How could we share mobility data on mobility? There’re different ways of doing it, and one of the way we could do this is through a Data Trust. Data Trust is a regulated jurisdiction we can put in place to transfer data and manage it.” - P4

While grappling with these new practices is challenging, some other organizations who had adopted technological pathways early on, spoke about the ease with which social-media platforms like Twitter allow them to easily disseminate information, resulting in direct multiplication of impact. A transit-advocate we spoke with started their organization on Twitter, and highlighted the advantages of Twitter in their work:

“So, we try to focus on that multiplication of impact by us recognizing what people are looking for, what groups are looking for, and seeing is there a way that we can do research in this area, so that we’re able to build up more, and get more information out there to more people. When we published our report, we did print off a few 100 copies but that’s, that’s such a, such an inefficient way to get information out there. So, we try to be very

online, make use of social media take advantage of opportunities to share information in a free way. So that it travels without us having to throw in a bunch of cost, you know, to do a bunch of marketing or that sort of thing, and then make it something that other groups can use to multiply it.” - P1

In this light, the work to meet the immediate needs of grant reporting is part of a strategy that aims to build high-impact structures capable of empowering citizens, and other mobility justice advocates to address some of the long-standing urban mobility injustices. Through building and utilizing tools like the learning management system, adopting digital communication in addition to offline communication, curating formal data practices, formalizing impact tracking and assessment, some mobility justice advocates are building the capacity and infrastructures that can propagate the necessary action needed to address the underlying infrastructural problems of urban mobility’s inequities.

4.3 The focus on being innovative in urban mobility limits the investment in well-known solutions and knowledge sharing.

While discussing the city’s tools and strategies for new mobility infrastructure in our conversations with mobility justice organizations, many of them brought up the frequent focus on innovation – new solutions designed for this region only. It is observed that novelty often brings greater attention and praise for the people spearheading these initiatives, leading to an ease in the procurement of funding for innovative projects. The groups we spoke with often expressed frustration at this focus on the “new” because they believe replicating a tried-and-tested solution that is already available brings about urban change faster and more effectively than coming up with a new experimental solution that has no guarantees. One of the transit-justice advocates we spoke to elaborates on this as follows:

“There is the lack of understanding that other cities have faced the same challenges. And they have found solutions to those challenges. Honestly, all of North America, but it’s really bad here, we are convinced that we are unique special snowflakes, and we must invent our own solutions. We can’t look to other places and learn from them. For example, originally, Toronto was going to build streetcar lines out in the suburbs, and they would branch out and they would cover a whole bunch of Etobicoke and Scarborough. And we had the technology for doing that. And we could have been built in such a way to be you know, high capacity, high speed, lots of options, it would not have to behave the way that the downtown streetcars do when they’re caught in traffic. But what the government asked us to do is to implement this new experimental Scarborough RT system.” - P8

Speaking of how novel experimental solutions often get stuck in the testing phase, do not get improved, and have detrimental funding consequences, another interviewee said:

“The Scarborough RT system was never upgraded, we’re still using the demo. And that’s why it’s broken. And that’s why it’s being shut down. Right, we could have upgraded... we should have done other things. But the idea of we invented this new system, and we want you to install the demo. And if you don’t, we’re going to cut your funding is just horrifying.” - P7

While most of the organizations we spoke with brought up this issue, they also outlined how in their own work, they focus on shifting this focus away from innovation for the general public through data that pulls from other regions and their mobility solutions. Many of these organizations prepare technical research reports that present this data in an accessible format and communicate the importance of borrowing solutions. One such organizer talks about this in more detail:

“We’ve also more recently, done co-presentations with city councilors within their own ward to talk about transit issues, talking about transit concepts. And in 2018, we released a research report called Mixed Signals³, which was a review of transit systems across North America and a bunch of different cities is comparing them with the Toronto Region and looking at the differences in the similarities and what we can learn from them.” - P1

Our participants also make use of communication strategies or “pop-ups” for knowledge sharing. Pop-ups are smaller scale demonstrations of mobility solutions aimed at gathering support prior to implementation in a neighborhood. In their work, mobility justice advocates find that these strategies are much more effective at shifting the focus away from innovation towards verified solutions, even if they’re borrowed. One of them spoke about the importance of weaving in storytelling with an emphasis on “temporary” pop-ups:

“Another important part of that plan is also the messaging, because like, we have this storytelling thing, where we create a key message for them that they can keep using. And, and that’s where, like, you know, we should keep saying that it’s temporary. And it’s a test, it’s a pilot, we are testing this so that, you know, the community is not thinking that oh, this is going to be a forever thing, and we have to fight this or whatever. It’s a, it’s a temporary thing that we are testing.” - P5

It has been noted by our participants that weaving in storytelling is often successful at gathering support from the neighbourhood residents. Another organizer talked about the importance of “visuals from other groups in the world” to increase knowledge sharing:

“We’re trying really hard not to be just a bunch of people with opinions, but people that are doing things and learning things and connecting with lots of people. We will often take visuals from other parts of the internet or other groups around the world. We’re not going to try and reinvent the wheel; we’re going to learn from other places and what they’ve done and take advantage of that.” - P12

The challenges in prioritizing novelty over practicality in designing mobility solutions are highly prevalent in the advocacy work our participants do, and the strategies they employ to shift focus away from novelty towards communicating about, and adopting, trusted approaches are still underdeveloped. While pop-ups are useful in bringing about some change, these proposals are often contested, and they need to be emphasized as “temporary”. In trying to solve these challenges, mobility justice advocates often get bound up in questions closely related to citizen rights and what the urban population deserves. The underlying questions of citizen rights with respect to urban mobility, focus on innovation and funding-related consequences of borrowing solutions, still remain largely unresolved.

4.4 Resistance to change constitutes a significant problem in bringing about a transformation in the current urban mobility landscape.

As outlined above, there are various contributors to urban mobility injustices, specifically, in the format of the information related to urban mobility planning, increased tech labor and requirements for mobility justice advocacy and the emphasis on seeming innovative. Envisioning solutions to address some of the above challenges with improved designs of communication channels and planning processes is possible. However, our interviews reveal an underlying systematic issue relating to the influence of “path-dependence” [60] in transportation technologies that may require more involved and thoughtful considerations from all stakeholders to bring about a transformative change to urban mobility. Our participants reveal that “path dependence” on the history of use and

³<https://coderedto.com/mixed-signals/>

adoption of mobility infrastructure and related technologies manifests in discussions around new transportation technologies and infrastructures. One interviewee talks about how the government, and planning experts often rely on assumptions in the design of suburban residential areas in Toronto where a majority of marginalized and immigrant populations reside:

“There is this perception that automatically in the suburbs, trips are going to be longer, we just assume that things are further apart in the suburbs. And so, trips are going to be longer. But looking in the GTHA (Greater Toronto and Hamilton Area), the data shows that actually fully 1/3rd of all trips in the GTHA are five kilometers or less. So, within a 20-minute bike ride or less, that’s 1/3rd of all trips that could be done by walking or biking very easily, but currently aren’t.” - P11

These beliefs and assumptions held by the people very clearly manifest in how our streets are designed, in people’s attitude regarding urban mobility and related public policies. When assessing plans and proposals for mobility infrastructures, Torontonians policies often dictate the prioritization of financial impact assessments, sometimes even over other human safety assessments. This is in line with Mackenzie et al’s observation about the entanglement of technological reasoning and economic reasoning. In their argument, they explain that the key factors for technical progress are future costs and future profits. Expressing frustrations over these procedures, one of our participants said:

“What was I think, certainly frustrating, though, is the fact that we know that separated bike lanes save lives, they save lives. Why is that not the beginning and end of the conversation, but it wasn’t enough. And so much of the data of emphasis was on the business impact. As we talked about before, the data showed it was very good business. And so, when the city then next piloted a bike lane in a different area, they didn’t need to do the business analysis. That was no longer the key central area.” - P2

Further elaborating on this issue, our participants talked about how these assumptions are not limited to just elected government representatives. They extend to urban citizens who are extremely “car-dependent”. Elected officials prefer to cater to these automobility needs based on the neighbourhoods they represent, and fear “political backlash” in case these needs are not met. This is even evident in advocacy work with parents and schools where issues with road safety and parking are prevalent, as one of our participants explains:

“It’s been interesting to me learning that grown up people do not want to be told what to do, like it’s even a surprise in some way, sometimes the principals and school staff have tried to speak to parents about driving and parking properly and it results in shouting matches.” - P14

Our participants observe that this fear of political backlash prevents the city’s prompt action and new street pop-ups. An organizer who works primarily with schools elaborated on their work with the city planners:

“First, they have to do like a whole analysis of what kind of school is good, or which neighborhood is good to do a pilot like this, because it shouldn’t be that challenging, but it should be a little challenging. If it’s too challenging for the residential community, then this project might just not take off. And, and the concept can just die, you know, like, no one would want to implement it, you know, or do more versions of it in the future. So, so yeah, they had to do like a whole analysis of finding the right school finding the right partners.” - P5

Many grassroots advocacy organizations that we interviewed recognize these barriers and are working with governments and resident communities to address this resistance to change

urban design, policy procedures and related technologies from the government and people. In addition to coming up with new forms of transport planning, zoning, congestion policies and traffic management, advocates are working directly with schools to build long-term working relationships to pilot transportation projects and co-create knowledge through collaborations with for children, their parents and other neighborhood residents. However, in doing so, they also recognize that this is a complicated issue that encompasses other questions concerning the politics of mobility, equity, urban accessibility and citizen rights that may not be solved through linear approaches to transportation planning and urban design. As we discuss in future sections, this challenge opens the door for conversations around a shift in policy considerations that rely on outdated assumptions and mobility data that is lacking in its representations of the lived realities of all people, but especially those that have been historically marginalized in urban mobilities.

5 DISCUSSION

Mimi Sheller writes that mobility justice requires us to “examine the influence of socio-political factors in the accessibility of movement and in the creation of uneven mobilities” [85]. In line with this, our findings highlight the role of contemporary digital technologies and data practices in the reproduction and reinforcement of mobility injustices. In the following sections, we identify path-dependence as a significant influencing factor in the lifecycle of transportation technologies. We argue that path-dependence is a systemic, underlying contributor to urban mobility injustices which should be taken into consideration by the HCI community. Drawing from our findings, we then discuss four tactics that offer opportunities for HCI research and design in HCI to unsettle path dependence, enabling movements towards mobility justice.

5.1 Path-Dependence as a Systemic Challenge for Mobility Justice

The challenges facing mobility justice advocates suggest that path-dependence is a common underlying problem. Path dependence refers to how “local, short-term contingencies” - e.g. market economics -, or “the path of its development” can “exercise lasting effects” on the kinds of socio-technical systems that may be developed in the future, [33, 60]. In the context of mobility, our findings echo prior work that demonstrates how developed economies like Toronto, get “locked-in” to carbon-intensive technologies, leading to self-reinforcing automobile cultures where car-dominant infrastructure design and daily individual movement routines revolve heavily around driving [30, 75]. Our advocates operate within these cultures and infrastructures, which continue to shape and constrain the range of mobility options we may consider, including the values and assumptions we draw upon to do so [83]. In their day-to-day activities, this appears up as backlash against “new” bike lanes and transport-inclusive streets or resistance to policies that seek to encourage marginalized modes of movement at the expense of more dominant ones [37]. Furthermore, the switching costs associated with some of our interviewees’ attempts at promoting behavioral change, like parents walking their kids to school instead of driving, contribute to the inertia [75] of current practices, and resistance to change.

The challenges disrupting path dependencies are rooted in their pervasiveness, manifesting not only at the technological level but also at institutional, infrastructural, and behavioral levels. This can make them difficult to even recognize in some cases, let alone undermine [92]. By operating at varying levels, path dependence constrains the actors and institutions looking for alternatives. Evident in the case of the broken Scarborough RT system, and noted by infrastructure studies more broadly [79], new technologies are developed upon an “installed base” of existing physical and social structures, aligning not only with the technical elements like roads and subway lines but also cultural discourses, business models, and regulations that continue to prioritise the production of fossil-fuel based car cultures. Compounding the challenge, fossil fuel lobbies wield considerable influence and

are well organized in contrast to the limited capacity and resources of mobility justice advocates [18]. They use this influence - comprising political, technological and behavioral forces - to contribute to the lock-in of otherwise diverse societies into particular structural configurations, making them resistant to re-orientation. As mobility justice advocates note, these structures perpetuate gradual, and frequently invisible harms, reinforcing transit inequalities, marginalizing vulnerable groups such as the poor, disabled, women, and children, who ultimately bear the brunt of environmental and social injustices associated with entrenched, carbon-intensive transportation systems [70].

The information systems that shape the politics of mobility thus do not exist in isolation, they operate within networks of technologies, physical, social and information elements, which then form a part of a broader knowledge infrastructure that provides the resources which both enable and constrain future research, design and knowledge work [43]. Some degree of path dependence is thus a necessary characteristic of large socio-technical systems, but scholars have repeatedly identified how this also creates a infrastructural biases [87], or a politics of infrastructure. For example, Murphy showed how sick building syndrome was (and remains) difficult to detect because toxicity benchmarks, including instrumentation, data standards, and relevant regulation are all tuned to legacy systems, and those who were first impacted were often less politically influential [67]. Similarly, the legacy maps, data and technologies that shape mobility planning and policy in Toronto today reflect a legacy of automobility, limiting and precluding pathways of investigation into alternatives for mobility. As our participants highlight, moving towards mobility justice is dependent upon undoing past systemic injustices, that, unless countered, current “business-as-usual” approaches will continue to reinforce [33].

Addressing the path dependencies of infrastructure will require more fundamental and transformative reorganizations to mobility-related design work. Design approaches for mobility information systems that continue to be top-down, expert-driven and divorced from the situated particularities of place - like the planners’ use of non-representative languages on transit maps or the narrow definitions of safety perceptions based on the number of cars - are unlikely to bring about any such change. These approaches are largely informed by traditions of scientific practice, that seek to abstract away the contentious values, biases, and valences embedded into socio-technical systems. This makes the very path dependencies that shape mobility politics difficult to recognize. HCI has well-documented critiques of such scientific and technical practices, and offered several useful pathways to overcome them [57, 61, 89]. Given that the City of Toronto, like many other cities around the world, are formulating their climate resilience strategies with a strong focus on transportation, we believe it presents an opportunity to engage with design work that can foreground and engage with the politics of transportation planning. In the following section, we draw on a range of tactics used by our advocates and put them in concert with these useful concepts from HCI to outline a series of design recommendations that can be used to identify and overcome path dependencies in mobility politics.

5.2 Overcoming Path Dependence

The current standards within transit planning, policies and systems that inform mobility planning are built upon knowledge and decision-making infrastructures that prioritize auto-mobility. Scott characterizes this as a form of “panoptic legibility”, that attempts to achieve a coherent ordering of movement across different settings [81], meeting the needs of bureaucratic structures surrounding city management and planning but failing to account for the diverse experience of urban life. In contrast, our findings demonstrate that to meet the needs of different types of movement, advocates draw on a variety of resources to carry out their advocacy work, ranging from city and municipal websites to consult local transit proposals, to online platforms like Twitter for expert transit knowledge and facts. These diverse technical, social and organizing practices attempt

to re-purpose existing decision-making processes to yield a form of “local legibility”, aimed at serving the heterogeneous mix of people and issues that create local context. HCI can play a role in supporting this form of infrastructure adaptation through bottom-up and participatory approaches to design [8, 65, 78]. Further, data systems for urban planning that enable interactions between government organizations, planners and the public can help amplify the perspectives and agency of local communities [21, 55].

As our findings suggest, many prevalent transit-related information practices currently serve to limit informed participation in mobility planning for the public, hinder advocacy efforts, and contribute to the path dependencies of current systems. This inaccessibility shows up in digital tools used by urban planners, including maps that do not support diverse language needs of residents in the region, or in homogenizing the situated aspects of the local neighborhoods they represent, promoting a sort of distance from their residents. This limits everyday transit users ability to identify and counter path-dependencies in transportation planning [43, 75]. HCI can help support alternative strategies to overcome these challenges. Digital artifacts that prioritise the use of visuals such as images and videos, and social media can help make transportation planning and related proposals more relevant and meaningful for the communities they work with. Situated mapping tools, interactive interfaces and data visualizations can support a variety of narratives, and enable community action [5, 44, 61]. Given that these forms of storytelling draw heavily on local understandings of place, HCI can engage placemaking researchers to help mobility justice advocacy with story-driven games, or other adaptive narratives within specific neighbourhoods [42, 65, 78]. More broadly, researchers can also help transportation planners to understand and work with the ways in which movement-related data can be more place-based [90].

One way to do that might be through the application of data infrastructure literacy principles to the transportation sector. Gray et al. define “data infrastructure literacy” as people’s competencies to not only understand and work with datasets but to also critically scrutinize and participate in the wider socio-technical infrastructures that produce these data [39]. As we see in this study, the release of transport datasets on official government websites and the use of data and data-related approaches primarily for funders and reporting on grants are encouraging, but in practice, they focus their advocacy primarily on increasing uptake of data, rather than on building capacities for using data-related approaches to empower advocacy work. Sometimes, this occurs because funders and communities don’t share the same values or priorities, and this variety cannot be effectively represented in a single information source. This leads to the prioritization of funder and donor requirements over local community needs. As an alternative, digital infrastructure literacy would involve the integration of various forms of information and different databases, offering a view of the different understandings of mobility, encouraging exploration of commonalities, differences, and unique boundaries. In line with prior HCI work, these forms of digital infrastructure practices allow information to be accessible by multiple kinds of people, with varying priorities and attachments to mobility planning [8].

In addition, work to unsettle the path dependencies in current mobility information practices would also benefit from the development of stronger working relationships between transportation planners and the public as well as different publics. In highlighting place-based and community-driven perspectives, and offering opportunities for groups with diverse concerns to create and engage with transportation information, alternative information infrastructures can shift priorities and values away from traditional models and enable new capacities across a shared collective. Engaging with such practices may help unsettle strict divisions between experts and the public. Doing so can also help transform these frameworks from *mechanisms* - supporting a particular form of activity [12] - to be *mediums* - supporting a broad range of activities, including developing the relationships and forms of solidarity across diverse groups necessary to challenge entrenched

systems [73]. Furthermore, HCI scholars have argued for incorporating “friction” and moments to pause, reflect and contest data and its socio-technical infrastructures as crucial design frames [9, 52, 73]. Such opportunities are necessary for the slow, careful, and collective labour of unpacking path dependencies.

Finally, design can also look for ways to support the adaptability of mobility information infrastructures. Through careful and continuous re-purposing and adaptation of existing systems and resources to align with contemporary challenges and needs of mobility justice, “older” knowledge infrastructures can also be deployed. Prior HCI literature has explored infrastructuring, an ongoing process that occurs across the range of a technology life-cycle – development, mediation and articulation, adaptation and re-design – to “align humans and non-humans (technologies, resources, spaces) for the emergence of new practices” [59], as a useful approach for this kind of engagement. Infrastructuring research rejects fixed or final notions regarding the form and potential of technological systems, but instead looks to the practices that enable their modification, adaptation, and appropriation in a particular context. These aspects of infrastructuring help us pay attention to the kinds of socio-political processes that create infrastructures, and the path dependencies that influence them. Furthermore, infrastructuring recognizes long-term shifts in the practices surrounding the use of infrastructures, which is important to identify and evaluate path dependence. HCI scholars can then help design technologies and interventions that can support adaptive forms of infrastructuring to help transition to a future beyond automobility in transportation.

6 LIMITATIONS AND FUTURE WORK

Finally, we would like to acknowledge the limitations of our study and chart opportunities for future work. It is important to recognize that the challenges discussed in this study are specific to the Greater Toronto Area (GTA). While mobility justice advocacy efforts share a common goal of driving change, the strategic approaches employed can vary across different geographic regions worldwide. This study highlights significant challenges faced by mobility justice advocates in the GTA, despite the region’s recognition for its robust local transit system (by North American standards) and sustainable transportation policies [22, 47]. However, it is essential to acknowledge that these challenges can differ for mobility justice advocates in other regions, where there might be alternative or additional complexities. Our interpretive approach was not intended to produce generalizable findings. Rather, we offer our contributions here as suggestive, as insights that may help guide or inspire similar work in other contexts.

Furthermore, we acknowledge that this work did not attempt to address the relevant privacy concerns related to mobility technologies. While our participants are deeply mindful of the privacy of their community residents, they do not themselves collect data on individual movement that would raise issues of privacy in their own work. They note that mobility justice and the work to address mobility injustice encompasses a broader scope beyond individual technologies. Mobility justice focuses on technologies that facilitate advocacy efforts and collective action, encompasses addressing systemic inequalities, promoting equitable access to transportation, and advocating for inclusive decision-making processes [85, 86]. Consequently, our discussion on this matter was confined to the wider challenges these advocates encounter in addressing mobility injustices. While individual privacy concerns remain a crucial aspect, they are but one facet of the larger mobility justice framework that seeks to address social, economic, and environmental disparities within transportation systems. Our goal with this research is to unfold unique mobility justice practices and suggest CSCW literature’s possible engagement through design.

7 CONCLUSION

As part of a global response to climate change, governments and grassroots organizations alike are trying to encourage forms of transportation that reduce emissions and make it easier for people to use public transit, bike and walk. However, this involves complex questions concerning the politics of mobility, equity, urban accessibility and citizen rights that can be difficult to solve. In this paper, we examined the role of digital technologies and data practices in the reproduction and reinforcement of mobility injustices. Through interviews with mobility justice advocates based primarily in the Greater Toronto area, we identify the need for HCI and CSCW researchers and designers to attend to the socio-political factors that influence mobility, develop tools and information sensitive to the diverse needs of urban communities, and recognize the interrelation of mobility with place and its residents. The work undertaken by the mobility justice advocates to empower local communities to shape the city's transportation policies and systems, highlights how path-dependence is a systemic cause of mobility injustices that should be addressed. Further, we echo arguments made in other research efforts in HCI and CSCW around infrastructuring to help redistribute agency, create new knowledge, and ultimately move beyond present cultures of automobility. We hope that this work will provide insights into the current state of mobility justice and help undermine some of the path dependencies that stand in the way of equitable access to transportation.

ACKNOWLEDGMENTS

We extend our sincere thanks to all the mobility justice advocates who generously shared their time, resources, and invaluable experiences during our interviews. Their insights were instrumental in shaping the depth and perspective of our research, and we are truly grateful for their meaningful contributions. Thank you to the Toronto Climate Observatory (TCO) for supporting this research, and our friends and colleagues at the TCO for their constructive feedback and enthusiasm. Finally, thank you to our anonymous reviewers for helping shape the final version of this paper.

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Received January 2023; revised July 2023; accepted Oct 2023